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RESEARCH ARTICLE

Robust Anti-Jamming Technique for UAV Data Collection in IoT Using Landing Platforms and RIS

ABDULLAH M. ALMASOUD^{ID}

Department of Electrical Engineering, Prince Sattam Bin Abdulaziz University, Al-Kharj 11942, Saudi Arabia

e-mail: am.almasoud@psau.edu.sa

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ABSTRACT Adopting Unmanned Aerial Vehicles (UAVs) in wireless networks provides promising methods to extend the coverage of the network. However, the uplink communications between the ground devices and the UAV are vulnerable to jamming. Although many works have exploited the maneuverability of the UAV to avoid the jammer, the UAV will not be effective when the jammer is impacting a large area. In this paper, the data collection problem in Internet of Things (IoT) using a UAV in the presence of a malicious jammer is investigated. A Reconfigurable Intelligent Surfaces (RIS) is adopted to help the ground IoT devices in reflecting their signals toward the UAV. Moreover, a novel network element, called Landing Platform (LP), is introduced to support a resilient data collection in the presence of high jamming power using network cables. A joint IoT devices clustering and UAV's trajectory planning is considered, where the problem is formulated mathematically. Then, the Ant Colony-based Algorithm for Joint Anti-jamming UAV Routing and IoT Devices Clustering (AC-JRC) is introduced to solve the optimization problem in an efficient way. Finally, the impact of the LPs and the performance of AC-JRC algorithm compared with other benchmarks is investigated. The objective is to minimize the weighted sum of the UAV's total traveled distance and the number of used LPs. It is shown that the solution of AC-JRC is within 9% of the optimal solution. Moreover, the AC-JRC algorithm can provide a solution better than the solutions of AC-kmeans and ACS algorithms by 17% and 61%, respectively.

INDEX TERMS Unmanned aerial vehicle (UAV), Internet of Things (IoT), reconfigurable intelligent surfaces (RIS), IRS, anti-jamming, landing platform, data collection, trajectory design, clustering, Ant Colony.

I. INTRODUCTION

The Unmanned Aerial Vehicle (UAV) is an emerging technology that allows automated operations for the aircrafts without humans onboard. The UAVs have been used mainly in the past for supporting the militaries' operations such as border surveillance and remote attacks to the hostiles. However, the UAVs have many civilian and commercial applications nowadays. For example, UAVs can assist in disaster warnings, search and rescue, carrying medical supplies for remote areas, infrastructure inspection, remote sensing, wildfire fighting, real-time monitoring and many other applications. Moreover, the UAVs can enable disruptive innovations and interesting

applications in the area of wireless communications and networking [1].

UAV-assisted wireless networks provide new methods and techniques in solving traditional problems in the networking domain such as data collection [2], data dissemination [3] and many other public and civil applications [1]. Although the UAV adoption in the wireless networks has a great impact in improving the network performance and services, the UAV imposes several challenges such as limited operation time, limited computing power and vulnerability to the jammers. Hence, we need to consider such challenges while designing and operating the UAV-assisted wireless networks.

The increasing demand for connecting many physical objects to the internet led to the proliferation of the Internet of Things (IoT) applications [4]. These physical objects or

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devices may be located outside the traditional communication coverage or require a dedicated and reliable method for backhauling their data. Hence, UAV-based data collection in the IoT has emerged as an effective and flexible solution in the wireless networks. Additionally, Reconfigurable Intelligent Surfaces (RISs), also known as Intelligent Reflecting Surfaces (IRSs), add a new important element to the wireless network architecture that improves the performance and provides a more flexible network architecture [5]. It is shown that the RIS can improve the bi-directional communication between two network elements such as the base station and the ground user when its location is optimized [6]. Adopting the UAV and the RIS in the IoT can significantly enhance the data collection process and provide more flexible and resilient network architecture for the network's users.

In [7], we proposed a data collection scheme for the IoT devices using a UAV in a presence of a jammer. considering the jammer location and its jamming signal radiation, the IoT devices are clustered and the UAV flying path is optimized accordingly. The objective is to minimize the UAV's arrival time. In [8], the RIS is deployed to assist the IoT device in sending its message to the UAV in a presence of a jamming signal. The goal is to maximize the average rate of the transmission from the IoT device to the UAV. This is achieved by jointly optimizing the IoT device's transmission power, UAV's flying path and RIS's beamforming.

Similar to [8], the authors in [9] proposed to adopt the RIS in the IoT in the presence of a jammer to assist the UAV in the data collection process. However, the aim was improving the energy efficiency by jointly optimizing the device power allocation and communication scheduling in addition to RIS's beamforming and the UAV's trajectory planning. In [10], a joint UAV trajectory and RIS beamforming design is proposed for anti-jamming UAV communication. The proposed design considers imperfect channel state information while multiple jammers are interfering with the ground IoT device's transmission to the UAV. The velocity of the UAV is also optimized in addition to the RIS's phase shift matrix to maximize the energy efficiency.

In [11], an RIS-assisted and anti-jamming system is introduced to help the UAV data collection process for the IoT applications. The RIS adapts its configuration based on the environment such that it enhances the desired signal and rejects the jamming signal. The authors proposed a deep reinforcement learning (DRL)-based approach to achieve these goals, where deep deterministic policy gradient (DDPG) and twin delayed DDPG (TD3) models are utilized. It is proposed in [12] to mount an RIS panel on the UAV in order to mitigate the jamming attack. A ground base station sends its messages to the UAV that conveys and relays them to a ground user. When the ground jammer starts its attack towards the UAV to disrupt a successful reception of the base station's messages, the UAV configures its RIS such that it successfully reflects these messages to the ground user.

On the other hand, an anti-eavesdropping communication method is introduced in [13] via a fixed on ground RIS and a flexible RIS mounted on a UAV. The UAV acts also as a friendly jammer to prevent the eavesdropper from receiving illegitimate information. The RIS helps in improving the secrecy by jamming the eavesdropper and supporting secure communications between the UAV and the ground user. In [14], a fixed UAV acts as a friendly jammer to confuse the eavesdropper while another UAV communicates with an on ground user. The RIS is also utilized to enhance the security of the communications between the transmitting UAV and the ground user. Additionally, the RIS cooperates with the fixed UAV in jamming the eavesdropper and protecting the legitimate user.

When the RIS is adopted in the network, it provides more flexibility for the UAV to collect data indirectly from the IoT device. The RIS is used in the proposed system to provide a more flexible and jamming-resilient UAV data collection scheme. Moreover, we introduce a new network component called a Landing Platform (LP) to be added to the network architecture. The LP is a platform designed in such a way that allows the UAV to safely land over it. Then, the UAV becomes physically attached to the LP, and they can exchange data. In addition, the LP is connected to the ground IoT device through a network cable. The IoT device physically connected to the LP can be selected as a cluster head for a set of nearby IoT devices, where the cluster head is located within their communication range. Hence, the IoT devices send their data to the cluster head over the unjammed channel between them. Then, the LP forwards the data sent to the cluster head to the UAV. On the other hand, we adopt the RIS in the network to help the UAV in data collection in presence of a jamming attack. The IoT devices are clustered carefully to avoid the jamming attack and minimize the UAV flying trajectory while collecting the data from the cluster heads.

The LPs should be distributed over the network area carefully such that the whole area becomes resilient to the jamming attack. This happens when an enough number of LPs are distributed evenly over the whole IoT devices' serving area. Given that the transmission range of the IoT device is R_{tx} , the LP forms a cell that can include all the IoT devices within the distance of R_{tx} . When an enough number of LPs are distributed evenly over the whole IoT devices' serving area, all IoT devices can be resilient to the jamming attack. This happens when each IoT device in the network is served by at least one LP as we will discuss this in Subsection II-A.

When clustering is adopted in the data collection process, the UAV may need to visit only a subset of the IoT devices instead of visiting all of them. This can simplify the data collection task and reduce the distance traveled by the UAV. Additionally, the formation of the clusters takes into consideration the location of the CHs since the flying trajectory depends on the locations of these selected CHs. The UAV starts and finishes its flight from predefined initial and final locations. Therefore, the clustering technique

should optimize the selection of these CHs mathematically and heuristically as we will elaborate on this in Section III and Section IV, respectively.

In this paper, we formulate a mathematical optimization to model our problem and be able to derive the optimal solution. Given certain constraints such as the UAV speed, the locations of the IoT devices and the LPs, maximum tolerated jamming power and other constraints as we will explain in Section III, the optimization problem aims in minimizing the weighted sum of the UAV's total traveled distance and the number of used LPs. Since deriving the optimal solution is computationally prohibitive when the scale of the problem is large, we propose the AC-JRC algorithm in Section IV to solve the problem in a more efficient way. This algorithm is based on the standard Ant Colony System Algorithm (ACS) while jointly considering the UAV routing and the IoT devices clustering. To demonstrate the effectiveness of this proposed algorithm and proposed network element (the LP), we study in Section V the impact of adopting the LPs, the RIS and utilizing the AC-JRC algorithm on the performance while considering several benchmarks.

Although there are several works adopting UAVs and RISs to support anti-jamming data collection [8], [9], [10], collecting data from ground users using the proposed schemes in the literature will not be possible when the attacker jams most or the whole area of the network. To the best of our knowledge, this is the first paper that introduces a new network infrastructure that we call the Landing Platform (LP), which mitigates the effect of high jamming power. By utilizing the proposed LPs, the RIS, Landing Platform-aware IoT devices clustering and UAV trajectory design, the UAV data collection becomes more resilient to the jamming attack even when the jamming affects a large area of the network.

A. MOTIVATIONS AND CONTRIBUTIONS

The ground sensors in the IoT are usually battery powered with limited charging or replacement opportunities. With the existence of a malicious jammer, the IoT devices performance may deteriorate due to the interference caused to the uplink channel to the UAV. Additionally, the jammer can block the UAV from collecting data from jammed IoT devices. Hence, the jammer may force some of the IoT devices to be re-clustered, as we will show later, and that can lead to a longer flying trajectory for the UAV and shorter operating time. Furthermore, the jammer in some scenarios can block entire, or at least many, IoT devices from sending their messages to the UAV when the jamming power is high. In this scenario, the jammer can cause a Denial-of-Service (DoS) attack to the network and prevent its normal operation. On the other hand, the UAV's operation time is limited due its reliance on a relatively small battery in general to reduce its weight. Furthermore, limited energy supply for the UAV restricts the total distance that can be traveled by the UAV. Hence, we are motivated to mitigate these challenges by optimizing the UAV's trajectory and the IoT devices clustering to minimize the flying distance while considering the jamming

attack. Moreover, we adopt the RIS and the LPs to improve the performance and the resilience to the jamming attack.

The main contributions of this paper include: 1) Introducing a new network component called Landing Platform (LP) to enhance jamming resilience, 2) Proposing a clustering scheme that considers the LPs and the RIS such that the cluster heads are resilient to the jamming attacks, 3) Formulating an optimization problem mathematically for UAV data collection in a presence of a jamming attack, 4) Proposing the AC-JRC algorithm for UAV data collection, which jointly considers the IoT device clustering and UAV routing and 5) Studying the performance of the AC-JRC algorithm and comparing it with several benchmarks.

The contributions of our paper can be described as follows:

- We propose a novel new network component called a Landing Platform (LP) to be added to the network architecture. The LP links the cluster head IoT devices to the UAV physically through a network cable to eliminate the jammed wireless medium when the jammer is active. The LP allows the UAV to land over it and receive the aggregated data from the cluster head of the associated cluster through a network cable. A certain number of LPs are distributed over the network area based on the communication range of the clustered IoT devices to their cluster heads. The LPs provide a resilient network architecture against the Denial-of-Service attack to the UAV-assisted wireless networks.
- The RIS is adopted in the network architecture to mitigate the jamming attack and/or minimizes the UAV's data collection flying trajectory. We propose a clustering scheme that is enhanced by the RIS, where the IoT device can be selected as a cluster head when its channel to the UAV through the RIS is the best among all its neighbors' channels to the UAV. Moreover, the UAV flying trajectory is influenced by the location of the RIS, where the UAV can collect data indirectly through the RIS when that minimizes its flying distance. Hence, we proposed an RIS-aware joint IoT devices clustering and UAV's trajectory planning to optimize the performance of the data collection process.
- After defining our problem, we formulate it mathematically as an Integer Linear Program (ILP). Solving this formulated problem provides an optimal RIS's configuration, LPs activation, IoT devices clustering and UAV path planning while considering a malicious jammer.
- We proposed an efficient Ant Colony-based Algorithm for Joint Anti-jamming UAV Routing and IoT Devices Clustering (AC-JRC) to solve our optimization problem. Inspired by the behavior of real ants, traditional Ant Colony Algorithm [15] solves the traveling salesman problem. However, we develop another Ant colony-based algorithm that is tailored to solve our problem. In addition to solving the optimal tour for the UAV, our proposed algorithm also clusters the IoT devices. Moreover, AC-JRC considers the jamming attack and

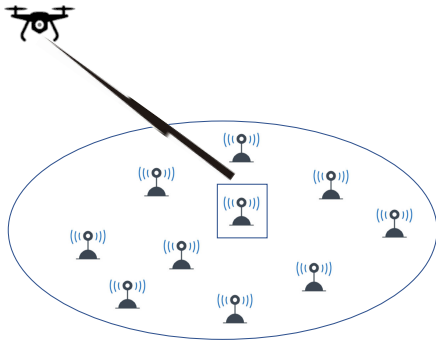


FIGURE 1. A cluster head IoT device transmits the cluster's data to the UAV.

takes the advantage of the RIS and LP when deriving the solution for our problem.

- The performance of the proposed scheme and algorithm are evaluated and compared with the exact optimal solution and other benchmarks. We investigated the impact of our proposed LPs, location of the RIS, jamming power and other factors on the performance of the UAV data collection.

The remainder of the paper is organized as follows. First, we describe the system model in Section II and formulate the problem mathematically in Section III. Then, we introduced our proposed algorithm in Section IV and presented the simulation results in Section V. Finally, we conclude our paper in Section VI. Throughout the paper, $\mathbb{C}^{W \times 1}$ represents $W \times 1$ complex vector, $*$, T and H denote the conjugate, transpose and Hermitian operation, respectively.

II. SYSTEM MODEL

We consider a UAV collecting data from a set of ground IoT devices, $\mathcal{I}$, in presence of a malicious jammer. The IoT devices are fixed and their locations are known to the UAV. Moreover, the location of the jammer is detected by the IoT devices using some accurate jammer localization techniques [16]. Then, they send the jammer's location to a central entity that pre-plans the optimal data collection path for the UAV. The IoT devices are distributed over a grid of size $x^{max} \times y^{max}$ m. In this grid, each IoT device is placed in one location, where the set of all IoT devices' locations is $\mathcal{L}$. The IoT devices are clustered into $|C|$ clusters, where C is the set of clusters with radii ϵ . Within each cluster, the IoT devices send their data to the Cluster Head (CH). Then, the UAV collects the aggregated data only from the CHs, as shown in Fig. 1, to expedite the process of data collection and to overcome the limited operation time for the UAV.

The UAV flies with altitude H m and speed S m/s. Let T be the set of time slots, where the data collection period is slotted into $|T|$ slots of duration δ . The UAV can fly from one location to another during one time slot given that the distance between these two locations is less than or equal to $S\delta$. Once the UAV reaches a data collection location, it polls the corresponding CH to send its data by the end of the scheduled time slot. The initial location of the UAV is I ,

TABLE 1. Symbols' descriptions [7].

Symbol	Description
	IoT device.
	Cluster head IoT device.
	UAV.
	Jammer.
	Routing path of the UAV.
	UAV's initial location.
	UAV's final location.
	Cluster of IoT devices.
	Area affected by the jammer.
	Landing Platform.
	RIS.

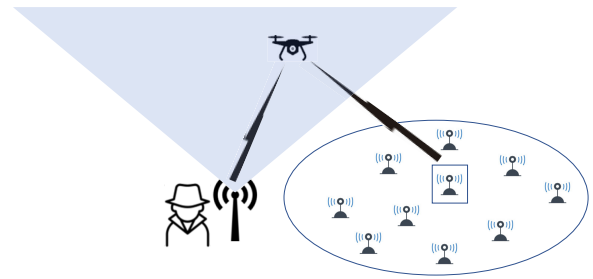


FIGURE 2. A jammer is attacking the uplink communication of the IoT devices.

where it flies to collect data from the IoT devices then lands to its final location F . We use in this paper the terms “IoT device i ” and “the IoT device located at location i ” interchangeably for the same intended meaning.

We assume that the jammer is attacking the uplink channels between the IoT devices and the UAV as shown in Fig. 2, where the transmission powers of the IoT devices and jammer are P and P_J , respectively. Let $\bar{\gamma}$ be the signal-to-interference-plus-noise ratio (SINR) threshold for the successful reception of the signal sent by the IoT device to the UAV. Hence, the UAV optimizes its data collection locations such that the SINR of the received signal is greater than or equals to $\bar{\gamma}$. For example, the UAV can optimize its data collection location to avoid the jammed area, as shown in Fig. 3. Moreover, the CHs selection is optimized such that the CHs can send their data without harmful interference when the UAV collects data from them, as shown in Fig. 4

On the other hand, we adopt Reconfigurable Intelligent Surfaces (RIS) in our system to provide a more flexible data collection scheme. Instead of collecting data directly from a CH, the RIS can reflect the CH' signal to the UAV. This can be helpful in avoiding data collection from a jammed area, as shown in Fig. 5. Furthermore, the RIS may provide better uplink channels for the CHs. In other words, the RIS may allow the UAV to collect data from locations that result in a shorter flying trajectory. Moreover, the RIS provides a resilient network architecture that can mitigate the jamming

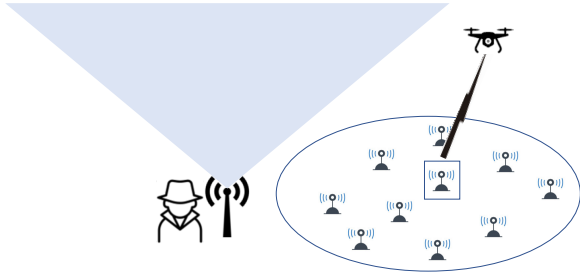


FIGURE 3. A jammer is trying to attack the uplink communication of the IoT devices, but the UAV adjusts its location to avoid the jamming.

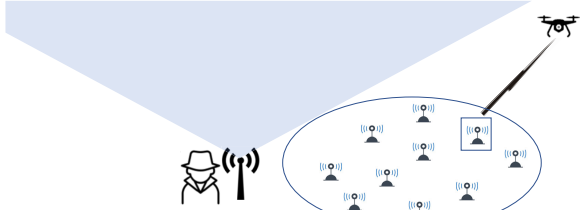


FIGURE 4. The jammer increases its jamming power to reach to attack the uplink channel, but the UAV adjusts its location and the CH is reselected to avoid the jamming.

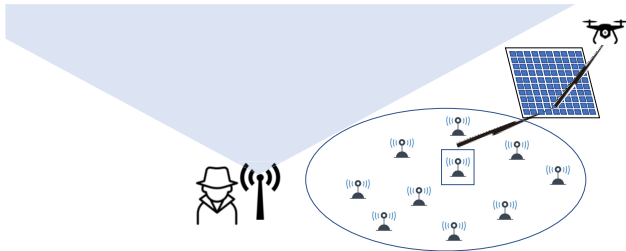


FIGURE 5. Optimizing the UAV location and utilizing the RIS to avoid the jamming area.



FIGURE 6. UAV Landing Platform (LP) architecture.

effect in some scenarios when the UAV adjusts its collection locations properly.

A. PROPOSED LANDING PLATFORM (LP)

We propose a novel network component, called Landing Platform (LP), for jamming-aware data collection schemes in UAV-assisted wireless networks. Fig. 6 shows the landing platform, which is always connected to an IoT device through a network cable. This platform allows the UAV to land over it and receives data from the IoT devices through a network cable. Therefore, the UAV can collect the data from the IoT device regardless of the presence of the jamming signal as shown in Fig. 7. The IoT device that is connected to the LP becomes a CH, and it collects the data from the cluster members through a channel other than the jammed channel used for the uplink communication to the UAV. Then, it forwards

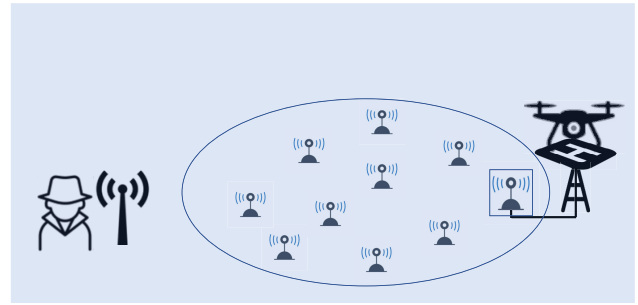


FIGURE 7. The UAV utilizes the LP to collect the cluster's data over a network cable rather than the wireless medium to avoid the jamming attack.

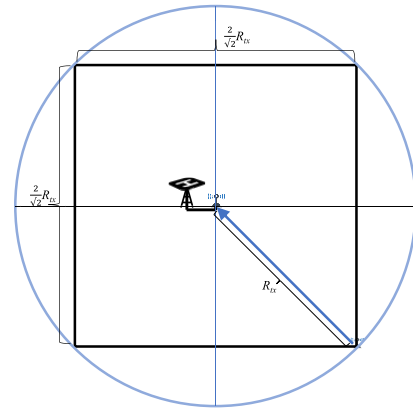


FIGURE 8. The LP cell includes all IoT devices within a distance of R_{tx} from the LP. The LP allows these IoT devices to send their data to the UAV indirectly through it when a jammer is attacking the uplink channel from the IoT device to the UAV.

the data corresponding to its cluster to the UAV through the LP.

Let the transmission range of the IoT device for its transmission to the LP is R_{tx} . Fig. 8 shows the LP cell (the black square) that includes an LP serving all IoT devices within that cell. Using the geometric principles, we can derive the dimension of the LP cell, which is $\frac{2}{\sqrt{2}}R_{tx} \times \frac{2}{\sqrt{2}}R_{tx}$. To make the IoT devices resilient to the jamming attack, we need to have a certain number of LP cells that cover the whole serving area for the IoT devices. The total number of needed LPs that are distributed evenly over the serving area and make the IoT devices resilient to the jamming attack is given by $\left\lceil \frac{x^{max}}{\frac{2R_{tx}}{\sqrt{2}}} \right\rceil \times \left\lceil \frac{y^{max}}{\frac{2R_{tx}}{\sqrt{2}}} \right\rceil$ where the number of LP on the x and y axes are $\left\lceil \frac{x^{max}}{\frac{2R_{tx}}{\sqrt{2}}} \right\rceil$ and $\left\lceil \frac{y^{max}}{\frac{2R_{tx}}{\sqrt{2}}} \right\rceil$, respectively. When the LP cells partially cover the serving area of the IoT devices, then the IoT devices within that covered area will be resilient to the jamming attack as shown in Fig. 9. However, Fig. 10 presents the scenario where the whole network is resilient to a jamming attack that affects all IoT devices since each IoT device is covered by at least one LP.

B. CHANNEL MODEL

We consider the uplink communication from the CHs IoT devices to the UAV. This communication can be direct from

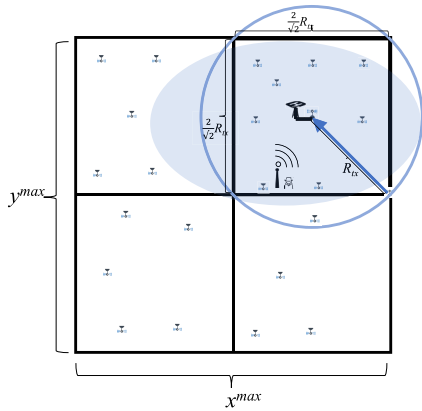


FIGURE 9. The area of the IoT devices are divided into 4 cells, and the upper right cell has an LP that serves all IoT devices within that cell against a jamming attack.

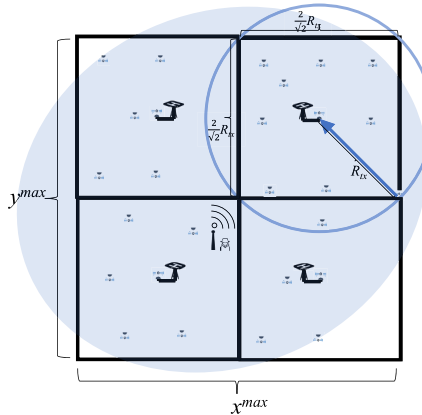


FIGURE 10. All IoT devices are served by four LPs to protect them against a jammer that transmits with a high power to affect all IoT devices.

the IoT to the UAV or indirect through the RIS, as shown in Fig. 2 and Fig. 5, respectively. Moreover, we consider the interference introduced by jammer, which also can be direct to the UAV or indirect through the RIS. The UAV and each IoT device are equipped with single omni-directional antennas. On the other hand, the RIS consists of a uniform planar array (UPA) with M elements. The number of vertical and horizontal elements in the RIS are M_z and M_x , respectively, where $M = M_x \times M_z$. Let Θ_{ik} be the diagonal phase-shift matrix for the RIS when a ground device (an IoT device or the jammer) located at location i is transmitting to the UAV located at location k through the RIS, where $\Theta_{ik} = \text{diag}\{e^{j\theta_1^{ik}}, e^{j\theta_2^{ik}}, \dots, e^{j\theta_M^{ik}}\}$. θ_m^{ik} is the phase shift of the m^{th} reflecting element when a ground device located at location i is transmitting to the UAV when it is located at location k , where $\theta_m^{ik} \in [0, 2\pi)$ and $m \in \{1, \dots, M\}$. The channel model used in our system can be described as follows:

1) DIRECT CHANNELS BETWEEN GROUND DEVICES AND THE UAV

The UAV in our system flies with a high altitude, where the blockages in the air are rare. Hence, we assume that the channels between the UAV and the ground device are

Line-of-Sight (LoS) channels [8]. Accordingly, the direct channel between an IoT device located at location i and the UAV when it is located at location k , $h_{ik}^{GU} \in \mathbb{C}^{M \times 1}$, is given by [8], [9]

$$h_{ik}^{GU} = \sqrt{\mu d_{ik}^{-2}} e^{-j \frac{2\pi d_{ik}}{\lambda}} \quad (1)$$

where $e^{-j \frac{2\pi d_{ik}}{\lambda}}$ represents the phase response, μd_{ik}^{-2} is the path loss, μ is the path loss at one meter reference distance, d_{ik} is the distance between location i and k and λ is the carrier wavelength. Similarly, the direct channel between the jammer and the UAV when it is located at location k is given by

$$h_k^{JU} = \sqrt{\mu d_{Jk}^{-2}} e^{-j \frac{2\pi d_{Jk}}{\lambda}} \quad (2)$$

where d_{Jk} is the distance between the jammer and location k .

2) CHANNELS BETWEEN GROUND DEVICES AND THE RIS

We calculate the channel between the IoT device, located at location i , and the RIS as follows:

$$h_i^{GR} = \sqrt{\mu d_{iR}^{-2}} e^{-j \frac{2\pi d_{iR}}{\lambda}} m_x^{iR} \otimes m_z^{iR} \quad (3)$$

where

$$m_x^{iR} = [1, e^{-j\alpha_x^{iR}}, \dots, e^{-j(M_x-1)\alpha_x^{iR}}]^T \quad (4)$$

$$m_z^{iR} = [1, e^{-j\alpha_z^{iR}}, \dots, e^{-j(M_z-1)\alpha_z^{iR}}]^T \quad (5)$$

$$\alpha_x^{iR} = \frac{2\pi d}{\lambda} \frac{l_i^z - l_R^z}{d_{iR}} \quad (6)$$

$$\alpha_z^{iR} = \frac{2\pi d}{\lambda} \frac{l_i^x - l_R^x}{d_{iR}} \quad (7)$$

where d_{iR} is the distance between the RIS and location i , d is the antenna separation, $\otimes$ denotes the Kronecker product. l_i^z and l_R^z are the z coordinates for location i and the RIS, respectively, and l_i^x and l_R^x are the x coordinates for location i and the RIS, respectively. Similarly, we calculate the channel between the jammer and the RIS as follows:

$$h^{JR} = \sqrt{\mu d_{JR}^{-2}} e^{-j \frac{2\pi d_{JR}}{\lambda}} m_x^{JR} \otimes m_z^{JR} \quad (8)$$

where

$$m_x^{JR} = [1, e^{-j\alpha_x^{JR}}, \dots, e^{-j(M_x-1)\alpha_x^{JR}}]^T \quad (9)$$

$$m_z^{JR} = [1, e^{-j\alpha_z^{JR}}, \dots, e^{-j(M_z-1)\alpha_z^{JR}}]^T \quad (10)$$

$$\alpha_x^{JR} = \frac{2\pi d}{\lambda} \frac{l_J^z - l_R^z}{d_{JR}} \quad (11)$$

$$\alpha_z^{JR} = \frac{2\pi d}{\lambda} \frac{l_J^x - l_R^x}{d_{JR}} \quad (12)$$

3) CHANNELS BETWEEN THE RIS AND UAV

The channel between the RIS and the UAV when it is located at location k , $h_k^{RU} \in \mathbb{C}^{M \times 1}$, is given by

$$h_k^{RU} = \sqrt{\mu d_{Rk}^{-2}} e^{-j \frac{2\pi d_{Rk}}{\lambda}} m_x^{Rk} \otimes m_z^{Rk} \quad (13)$$

where

$$m_x^{Rk} = [1, e^{-j\alpha_x^{Rk}}, \dots, e^{-j(M_x-1)\alpha_x^{Rk}}]^T \quad (14)$$

$$m_z^{Rk} = [1, e^{-j\alpha_z^{Rk}}, \dots, e^{-j(M_z-1)\alpha_z^{Rk}}]^T \quad (15)$$

$$\alpha_x^{Rk} = \frac{2\pi d}{\lambda} \frac{H - l_R^z}{d_{Rk}} \quad (16)$$

$$\alpha_z^{Rk} = \frac{2\pi d}{\lambda} \frac{l_k^x - l_R^x}{d_{Rk}} \quad (17)$$

d_{Rk} is the distance between the RIS and the UAV when it is located at location k , and l_k^x is the x coordinate of location k .

4) CHANNELS BETWEEN GROUND DEVICES AND THE UAV THOUGH THE RIS

The channel between the IoT device located at location i and the UAV located at location k , when the communication goes through the RIS, is calculated as follows:

$$h_{ik}^{GRU} = (h_i^{GR})^H \Theta_{ik} h_k^{RU} \quad (18)$$

We assume that the UAV sends information to the RIS about the location of the IoT device that intends to transmit to the UAV through the RIS. Then, the RIS configures its reflecting elements accordingly. This procedure is repeated when the UAV plans to collect data from another IoT device in order to reconfigure the RIS's reflecting elements according to the locations of the UAV and IoT device.

Let h_{ik}^{GRU} be the channel between location i and k though the RIS, which is given by

$$h_{ik}^{GRU} = (h_i^{GR})^H \Theta_{ik} h_k^{RU} \quad (19)$$

From (3) and (13), the optimal configuration for the reflecting elements that maximizes the instantaneous SINR is given when [17]

$$\Theta_{ik} = \text{diag} \left\{ \left[\frac{(h_i^{GR})^* h_k^{RU}}{\sqrt{\mu d_{iR}^{-2}} \sqrt{\mu d_{Rk}^{-2}}} \right]^{-1} \right\} \quad (20)$$

Thus, the optimal configuration for the RIS's elements results in $h_{ik}^{GRU} = \sqrt{\mu d_{iR}^{-2}} \sqrt{\mu d_{Rk}^{-2}}$ where the SINR for the signal transmitted by a ground device located at location i to the UAV when it is located at location k is calculated as follows:

$$\gamma_{ik} = \frac{P |h_{ik}^{GU} + (h_i^{GR})^H \Theta_{ik} h_k^{RU}|^2}{P_J |h_k^{JU} + (h^J)^H \Theta_{ik} h_k^{RU}|^2 + \sigma^2} \quad (21)$$

where σ^2 is the power of the additive white Gaussian noise (AWGN).

C. ILLUSTRATIVE SCENARIOS

Fig. 11 shows an illustrative scenario for our proposed system, and Table 1 describes the meaning of the used icons. In step 1, the UAV flies from its initial location and collects the sensing data from the two clusters in step 2 and 3. Two

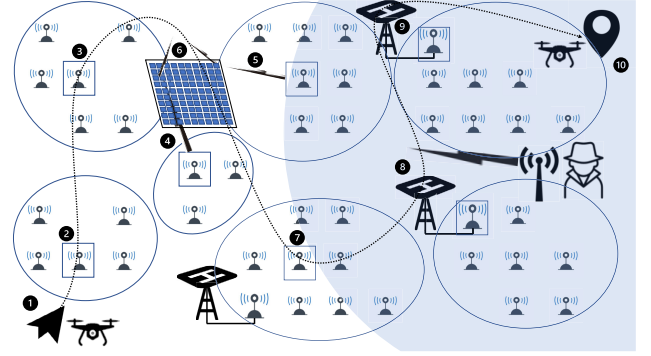


FIGURE 11. An illustrative scenario for UAV data collection with LPs and RIS adoption in a presence of a jamming attack.

TABLE 2. List of notations.

Symbol	Description
$\mathcal{L}$	Set of the locations on the 2D map.
$\mathcal{I}$	Set of the locations of the IoT devices.
$\mathcal{C}$	Set of IoT devices clusters.
$\mathcal{T}$	Set of time slots.
$X_{ij}(t)$	Binary variable equals one only when the UAV moves with altitude H from location i to location j during slot t .
x^{max}	Maximum value for x coordinate.
y^{max}	Maximum value for y coordinate.
CH	Cluster head.
ϵ	Cluster radius.
U_i^c	Binary variable equals one only when IoT device i is associated with cluster c .
H_i^c	Binary variable equals one only when the device i is the CH of cluster c .
$T(i, q, v, t)$	Binary variable equals one only when IoT device i is transmitting to the UAV flying from location q to v during slot t .
$F(i, t)$	Binary variable equals one only when the UAV lands to the platform of IoT devices i during slot t .
l_i	Location of IoT device i in the three dimensional space.
l_J	Location of the jammer in the three dimensional space.
I, F	Initial and final location of the UAV, respectively.
μ	Channel power at the reference distance $d_0 = 1$ m.
σ^2	Power of the additive white Gaussian noise (AWGN).
H	Height of the UAV.
$\hat{i}$	The location above IoT device i by H meters.
$\bar{\gamma}$	Signal-to-Noise Plus Interference Ratio (SINR) threshold.
γ_{ik}	SINR for the signal transmitted by IoT devices i and received by the UAV while it is hovering at location k .
P	Transmission power of the uplink transmission from the IoT device to the UAV.
P_J	Transmission power of the jammer.
S	Speed of the UAV.
δ	Time slot duration in seconds.
d_{ij}	Distance between location i and j in meters.
d_{Jk}	Distance between the jammer and the UAV when the UAV is at location k .
f_i	A parameter equals one only when the IoT device located at location i is equipped with a landing platform and zero otherwise.

CHs in step 4 and 5 send the data of their corresponding cluster to the RIS, which reflects the signals to the UAV in step 6. The RIS here contributes to minimizing the length of the flying trajectory because the UAV is not obligated to visit both CHs. Moreover, the RIS provides the UAV more flexibility to avoid collecting data from a jammed area in 5. Then, the UAV continues collecting data in step 7 and enters the jammed area. Interestingly, the UAV lands over the LPs in step 8 and 9 since it is impossible to collect the data from the jammed clusters using the wireless channels. Finally, the UAV heads to its final location and lands there in step 10.

III. PROBLEM FORMULATION

In this section, we formulate our problem while considering the UAV trajectory planning, IoT devices clustering, SINR requirements, jamming effect, landing platforms locations. We classify most of these constraints into three main categories as follows:

A. UAV TRAJECTORY PLANNING CONSTRAINTS

Let $X_{ij}(t)$ be a binary variable that equals one only when the UAV moves with altitude H from location i to location j during slot t . The UAV can move between only two locations during any given time slot, i.e.,

$$\sum_{i \in \mathcal{L}} \sum_{j \in \mathcal{L}} X_{ij}(t) \leq 1, \forall t \in T. \quad (22)$$

The UAV moves between two locations at most one time during all time slots. Therefore,

$$\sum_{t=1}^{|T|} X_{ij}(t) \leq 1, \forall i, j \in \mathcal{L}. \quad (23)$$

When the UAV is on location i , it moves from i only to at most one other location during all time slots, i.e.,

$$\sum_{t=1}^{|T|} \sum_{j \in \mathcal{L}} X_{ij}(t) \leq 1, \forall i \in \mathcal{L}. \quad (24)$$

Additionally, the UAV should not move back to its current location again during any time slot. Hence, we have

$$\sum_{t=1}^{|T|} X_{ii}(t) = 0, \forall i \in \mathcal{L}. \quad (25)$$

On the other hand, the UAV is initially located at location I , and it lands on its final location F after accomplishing the data collection task. Hence, we add these four constraints:

$$\sum_{j \in \mathcal{L}} X_{Ij}(1) = 1. \quad (26)$$

$$\sum_{t=1}^{|T|} \sum_{i \in \mathcal{L}} X_{iI}(t) = 0. \quad (27)$$

$$\sum_{t=1}^{|T|} \sum_{i \in \mathcal{L}} X_{iF}(t) = 1. \quad (28)$$

and

$$\sum_{t=1}^{|T|} \sum_{j \in \mathcal{L}} X_{Fj}(t) = 0. \quad (29)$$

When the UAV goes to a location j , other than I and F , it must leave that location in the next time slot. Accordingly, we have

$$\sum_{i \in \mathcal{L}} X_{ij}(t) - \sum_{k \in \mathcal{L}} X_{jk}(t+1) = 0, \quad j \in \mathcal{L} \setminus (I \cup F), \quad 1 \leq t < |T|. \quad (30)$$

The UAV movement from a location to another is physically limited by the speed of the UAV and the duration of the time slot. In other words, the maximum distance that can be traveled during each time slot is $S \delta$ meters. Therefore, we have

$$X_{ij}(t) d_{ij} \leq S \delta, \quad \forall i, j \in \mathcal{L}, \forall t \in T. \quad (31)$$

where d_{ij} is the distance between location i and j .

B. IoT DEVICES CLUSTERING CONSTRAINTS

Let U_i^c be a binary variable equals one only when the IoT device i is associated with cluster c and H_i^c be a binary variable equals one only when the device i is the CH of cluster c . First, the IoT device is associated to only one cluster, i.e.,

$$\sum_{c=1}^{|\mathcal{C}|} U_i^c = 1, \quad \forall i \in \mathcal{I}. \quad (32)$$

For each cluster, only one IoT device becomes a CH. Hence, we have

$$\sum_{i=1}^{|\mathcal{I}|} H_i^c \leq 1, \quad \forall c \in \mathcal{C}. \quad (33)$$

Moreover, the IoT device can be a CH but only to one cluster at most, i.e.,

$$\sum_{c=1}^{|\mathcal{C}|} H_i^c \leq 1, \quad \forall i \in \mathcal{I}. \quad (34)$$

When a cluster c is associated with an IoT device, then cluster c must have a CH. We express this constraint by

$$\frac{\sum_{j=1}^{|\mathcal{I}|} U_j^c}{|\mathcal{I}|} \leq \sum_{i=1}^{|\mathcal{I}|} H_i^c, \quad \forall c \in \mathcal{C}. \quad (35)$$

Similarly, an IoT device cannot be a cluster head to an empty cluster of IoT devices, i.e.,

$$\sum_{i=1}^{|\mathcal{I}|} H_i^c \leq \sum_{i=1}^{|\mathcal{I}|} U_i^c, \quad \forall c \in \mathcal{C}. \quad (36)$$

Within each cluster, the distance between the CH and any IoT device that is a member of that cluster is less than or equal to the cluster radius, ϵ . Then, we express this constraint by

$$U_i^c H_h^c d_{ih} \leq \epsilon, \quad \forall i, h \in \mathcal{I}, c \in \mathcal{C}. \quad (37)$$

C. DATA COLLECTION RELATED CONSTRAINTS

Let $T(i, q, v, t)$ be a binary variable that equals one only when the IoT device i is transmitting to the UAV flying from location q to v during slot t . First, the SINR of the signal, transmitted by the IoT device i and received by the UAV at the end of slot t when it is located at location v , must be greater than or equals to a threshold $\bar{\gamma}$, i.e.,

$$T(i, q, v, t) \bar{\gamma} \leq \gamma_{iv}, \quad \forall i \in \mathcal{I}, \forall q, v \in \mathcal{L}, \forall t \in T. \quad (38)$$

Additionally, the UAV is required to reach location v by the end of slot t to collect the data from the IoT device i when $T(i, q, v, t) = 1$. Hence, we have

$$T(i, q, v, t) \leq X_{qv}(t), \quad \forall i \in \mathcal{I}, \forall q, v \in \mathcal{L}, \forall t \in T. \quad (39)$$

Let $F(i, t)$ be a binary variable equals one only when the UAV lands to the LP connected to the IoT devices i during slot t . $F(i, t) \neq 1, \forall t \in T$ when $f_i = 0$, where f_i is a parameter equals one only when IoT device i is equipped with a landing platform and zero otherwise. Therefore, we have the following constraint:

$$F(i, t) \leq f_i, \quad \forall i \in \mathcal{I}, \forall t \in T. \quad (40)$$

When the IoT device is a CH, then it must either transmit its cluster data directly to the UAV or through an LP. This constraint is expressed mathematically by

$$\sum_{\forall c \in \mathcal{C}} H_i^c = \sum_{t=1}^{|T|} [F(i, t) + \sum_{\forall q \in \mathcal{L}} \sum_{\forall v \in \mathcal{L}} T(i, q, v, t)], \quad \forall i \in \mathcal{I}. \quad (41)$$

and

$$\sum_{t=1}^{|T|} [F(i, t) + \sum_{\forall q \in \mathcal{L}} \sum_{\forall v \in \mathcal{L}} T(i, q, v, t)] \leq 1, \quad \forall i \in \mathcal{I}. \quad (42)$$

When the UAV collects data from a CH through an LP, it needs to reach the location above that LP before landing over it. Therefore, we add this constraint:

$$F(i, t) \leq \sum_{\forall q \in \mathcal{L}} X_{qi}^c(t), \quad \forall i \in \mathcal{I}, \forall t \in T. \quad (43)$$

where $\hat{i}$ indicates the UAV location above IoT device i with altitude H .

D. OPTIMIZATION PROBLEM

The objective of the optimization problem is to minimize the weighted sum of the total traveled distance by the UAV and the cost of landing to the LPs. Accordingly, we can formulate our optimization problem as follows:

$$\begin{aligned} & \underset{\substack{X_{ij}(t), U_i^c, H_i^c, \\ F(i, t), \\ T(i, q, v, t)}}}{\text{minimize}} \sum_{t=1}^{|T|} \sum_{\forall q \in \mathcal{L}} \sum_{\forall v \in \mathcal{L}} [X_{qv}(t) d_{qv} \Lambda_1 + \sum_{i=1}^{|I|} F(i, t) \Lambda_2] \end{aligned} \quad (44)$$

subject to : (22 – 43)

$$\begin{aligned} & X_{qr}(t), U_i^c, H_i^c, T(i, q, v, t), F(i, t) \in \{0, 1\} \\ & \forall q, r \in \mathcal{L}, \quad \forall i \in \mathcal{I}, \forall c \in \mathcal{C}, \forall t \in T. \end{aligned} \quad (45)$$

where Λ_1 and Λ_2 are weighting factors for the objective function. In this optimization problem, we have two objective functions combined in one single objective function using the weighted sum method. In other words, we are minimizing the weighted sum of the total traveled distance by the UAV and the number of used LPs during the UAV flight. The reason for

minimizing the number of used LPs is that the UAV will need to hover over the LP, land carefully over the LP, exchange data with the CH associated with that LP and fly again to continue its flight. Hence, minimizing the number of used LPs will eliminate the delay caused for data collection using these platforms. Accordingly, the UAV expedites the data collection process by minimizing the traveled distance and the number of used LP. However, the LP should be used when it is not possible to collect the data wirelessly from the IoT devices associated with this LP due to the jamming attack.

The weighting factor Λ_1 and Λ_2 are used to weight the total traveled distance and the number of used LPs, respectively. The selections for Λ_1 and Λ_2 are based on the fact that the UAV should not use an LP if the uplink channel between the CH associated with this LP and the UAV is not under a jamming attack. Hence, the UAV avoids the delay caused by data collection using the LPs if there is no jamming. However, the UAV needs to utilize the LP when the CH associated with this LP suffers from the jamming. Therefore, we select any values for Λ_1 and Λ_2 such that $\Lambda_1 \ll \Lambda_2$ to avoid using the LPs unless they are needed. By setting $\Lambda_1 \ll \Lambda_2$, we minimize the total traveled distance by the UAV and utilizing the LP only when collecting the data from the CHs is not possible without the LPs.

E. LINEARIZATION

The formulated optimization problem is nonlinear due to the product of two binary variables, U_i^c and H_h^c , in constraint (37). To facilitate solving our optimization problem, we linearize the constraint (37) simply by introducing a new binary variable $Y_{ih}^c = U_i^c H_h^c$. Then, we replace constraint (37) by the following:

$$Y_{ih}^c \leq U_i^c, \quad \forall i, h \in \mathcal{I}, c \in \mathcal{C}. \quad (46)$$

$$Y_{ih}^c \leq H_h^c, \quad \forall i, h \in \mathcal{I}, c \in \mathcal{C}. \quad (47)$$

$$U_i^c + H_h^c - 1 \leq Y_{ih}^c, \quad \forall i, h \in \mathcal{I}, c \in \mathcal{C}. \quad (48)$$

and

$$Y_{ih}^c d_{ih} \leq \epsilon, \quad \forall i, h \in \mathcal{I}, c \in \mathcal{C}. \quad (49)$$

After this linearization, the reformulated optimization problem is given by

$$\begin{aligned} & \underset{\substack{X_{ij}(t), U_i^c, H_i^c, \\ Y_{ih}^c, F(i, t), \\ T(i, q, v, t)}}}{\text{minimize}} \sum_{t=1}^{|T|} \sum_{\forall q \in \mathcal{L}} \sum_{\forall v \in \mathcal{L}} [X_{qv}(t) d_{qv} \Lambda_1 + \sum_{i=1}^{|I|} F(i, t) \Lambda_2] \end{aligned} \quad (50)$$

subject to : (22 – 36), (38 – 43), (46 – 49)

$$\begin{aligned} & X_{qr}(t), U_i^c, H_i^c, T(i, q, v, t), F(i, t), Y_{ih}^c \in \{0, 1\} \\ & \forall q, r \in \mathcal{L}, \forall i, h \in \mathcal{I}, \forall c \in \mathcal{C}, \forall t \in T. \end{aligned} \quad (51)$$

This optimization problem is in the form of Integer Linear Program (ILP), which can be solved using any standard ILP solvers.

IV. PROBLEM SOLUTION

In Section III, the decision problem of the ILP program is NP-Complete [18]. Hence, the solution of the optimization problem is intractable when the number of variables is very large. To mitigate this challenge, we propose a modified version of Ant Colony System [15], [19] to solve our problem in a more efficient method. In the following, we provide a brief background about Ant Colony System (ACS) Algorithm. Then we describe our proposed Ant Colony-based Algorithm for Joint Anti-jamming UAV Routing and IoT Devices Clustering (AC-JRC).

A. ANT COLONY SYSTEM BACKGROUND

The ants in nature search for the shortest path to a food source without using visual cues. This phenomenon is possible due to the fact that the ants deposit pheromone along the path from their nest to the source of food. Then, the following ants discover the favorite path by going through the path with higher pheromone concentration. Interestingly, the higher pheromone concentration indicates a higher probability of finding a food source. When this source is no longer interesting for the ants, more ants will avoid going to this source. Consequently, less pheromone concentration will be on the path to that source since the pheromone evaporates with time. Hence, more ants will search for other food sources to discover better opportunities.

Ant Colony System (ACS) for solving the Traveling Salesman Problem (TSP) was introduced in [15] and [19], which is described briefly as follows. Inspired by nature, ACS algorithm iteratively searches for the best solution for the TSP problem using K ants. Each ant visits all cities and returns to its first cities. Let $J_k(r)$ be the set of yet not visited nodes (or cities in TSP problem) by ant k when it is located at node r . Given that ant k is currently at node r , it goes to node s with the following probability:

$$s = \begin{cases} \arg \max_{u \in J_k(r)} \{[\tau(r, u)] [\eta(r, u)]^\zeta\}, & \text{if } q \leq q_0 \\ S, & \text{otherwise.} \end{cases} \quad (52)$$

where $0 \leq q_0 \leq 1$. $\tau(r, u)$ is a real value corresponding to the pheromone left by ant k on the edge between node r and u . The value of $\tau(r, u)$ may decrease with time due to pheromone evaporation or may increase with time due to more phlegmon left by ants passing between node r and node u . $\eta(r, u)$ is a heuristic function, which is defined as the inverse of the distance between node r and s , and ζ is a weighting factor used to control the importance of the heuristic function, and S is a random variable selected base on the probability given by

$$p_k(r, s) = \begin{cases} \frac{[\tau(r, s)] [\eta(r, s)]^\zeta}{\sum_{u \in J_k(r)} [\tau(r, u)] [\eta(r, u)]^\zeta}, & \text{if } s \in J_k(r) \\ 0, & \text{otherwise.} \end{cases} \quad (53)$$

From (53), the ant prefers the next node to be visited based on the distance to that node and the amount of pheromone on the edge. During the ant's movement from node r to s , it applies pheromone along the edge between r to s . This step is called the local updating rule in the ACS algorithm. Once all ants found their paths to the food sources, more pheromone is applied to all edges belonging to the shortest resulting path among all ants' paths. Likewise, this step is called the global updating rule in the ACS algorithm. ACS algorithm iterates for a certain number of times, while the ants continue applying more pheromone during their tours. Hence, the ants tend to find better solutions during each iteration until no better solution is found or the ACS algorithm terminates.

B. ANT COLONY-BASED ALGORITHM FOR JOINT ANTI-JAMMING UAV ROUTING AND IoT DEVICES CLUSTERING (AC-JRC)

In this subsection, we propose our Ant Colony-based Algorithm for Joint Anti-jamming UAV Routing and IoT Devices Clustering (AC-JRC). It is shown in [15] that ACS algorithm solves the TSP problem while outperforming other nature-inspired algorithms. The ACS algorithm can be used to solve data collection problems in the IoT using the UAV when we consider the IoT devices as cities and the UAV as the ant that visits the IoT nodes. By analogy with an ant searching for the shortest path to a food source, the UAV searches for the shortest path from its initial location to the final destination.

However, the classical ACS algorithm cannot be adopted to solve our problem since our goals are different from just solving the TSP problem. Compared with ACS algorithm, our proposed algorithm is different due to the following:

- Instead of collecting data from all IoT devices, they are clustered and the UAV collects data only from the CHs, i.e., the ants in our algorithm visits only a subset of the IoT devices. Since we assumed in our problem that the radius of the cluster is ϵ , each IoT device that is not a CH should be located within ϵ m from the node visited by the ant.
- There is a restriction on the CH selection due to the existence of the jamming signal. All IoT devices that cannot upload their clusters' data to the UAV due to the jamming attack cannot be a CH except the IoT device connected to LPs. Hence, jamming signal radiation and the availability of the LPs play important rules in deciding the possibility of selecting an IoT as a CH. Accordingly, the ants in our algorithm are restricted on the selection of the nodes to be visited.
- Classical ACS algorithm is trying to find the shortest path to the destination while visiting all nodes. In contrast, the ants in our algorithm consider additional cost while visiting some nodes under jamming attack and connected to LPs. This extra cost is corresponding to the landing to the LPs, where landing to the LPs consumes an extra time and prolongs the flying duration. Hence, the local and global update for the pheromone over the

edges are affected by the ants' decision to visit some nodes under jamming attack and connect to the LPs.

- In the ACS algorithm, all nodes are visited, and the ant returns to its first visited node. By contrast, all ants in our algorithm start at location I and finish their tours at location F while visiting only a subset of the nodes. Additionally, the ant in our algorithm does not visit all nodes because they consider node clustering. Hence, our algorithm can be considered as a special case for the TSP problem where the first and last node to be visited are known in advance and only a subset of the cities are visited.
- Since we adopt the RIS in the system, the UAV may visit the RIS to collect the reflected signal sent by a CH. Therefore, the ant in our algorithm may visit the RIS instead of visiting a certain node nearby the RIS while considering that node as a visited node.

Algorithm 1 shows all the required steps of our Ant Colony-based AC-JRC algorithm. In lines 1-3, the algorithm starts initializing parameters and sets that will be used during its iterations, where CCH is the set of Candidate Cluster Heads, and $\bar{F}_i$ is a binary parameter that is equal to one only when IoT device i is attacked by the jammer and it is connected to an LP. In lines 4-11, AC-JRC algorithm derives the CCH set and the values of the parameters $\bar{F}_i, \forall i \in \mathcal{I}$. Hence, the ants in AC-JRC algorithm can consider a node to be a CH only when it is in the CCH set.

In lines 12-43, K ants derive their paths from I to F , where the path of ant k is denoted by Tour_k . Each $\text{Tour}_k, 1 \leq k \leq K$, may have different set of CHs. In line 13, the set CCH is copied in $\bar{\text{CCH}}$ in order to modify $\bar{\text{CCH}}$ while keeping the original set CCH. Then, node I is added to the tour of ant k in step 16 as the first node in the tour. The set of visited nodes by ant k , $\text{visited}(k)$, and the set of IoT devices to be visited, IoT_list , are initialized in lines 15-16. In line 17, the first CH for ant k , r_k , is selected randomly from the cluster of the candidate CHs, $\bar{\text{CCH}}$. Then, r_k is removed from $\bar{\text{CCH}}$ in line 18 to avoid selecting it again as a CH by the same ant. Moreover, the set of to be visited IoT devices by ant k , $J_k(r_k)$, $\text{visited}(k)$ and Tour_k are updated in steps 21-32 after selecting the first CH by ant k .

Ant k continues its mission of deciding the CHs to be visited and building its tour's path, Tour_k , as described in lines 22-42. In line 22, the list of IoT devices is reordered randomly to allow each ant to discover different solutions by adding the IoT devices to their tours in different orders. In lines 24-30, an IoT device is selected and it is included to the cluster of CH r_k only when it is not visited yet and its distance to r_k is not greater than ϵ . Otherwise, this IoT device is not included in the cluster of r_k . Ant k continues searching for other IoT devices satisfying the condition in line 25 to add them to r_k 's cluster.

Ant k selects the next CH to be visited, s_k , using equation (52) and (53). s_k is selected from the set $[(\mathcal{I} \setminus \text{visited}(k)) \cap$

Algorithm 1 Ant Colony-Based Algorithm for Joint Anti-Jamming UAV Routing and IoT Devices Clustering (AC-JRC)

```

1: Set  $\tau(r, s) = \tau_0, \tau(I, s) = \tau_0, \tau(s, I) = 0, \tau(s, F) = \tau_0$ , and
    $\tau(F, s) = 0, \forall r, s \in \mathcal{I}$ .
2:  $\text{Counter} = 1$  and  $\text{Tour}^* = \phi$ .
3:  $\text{CCH} = \phi, \bar{F}_i = 0, \forall i \in \mathcal{I}$ .
4: for ( $i = 1$  to  $|\mathcal{I}|$ ) do
5:   if ( $\gamma \leq \gamma_{ii}^*$ ) then
6:      $\text{CCH} = \text{CCH} \cup i$ .
7:   else if  $f_i = 1$  then
8:      $\text{CCH} = \text{CCH} \cup i$ .
9:      $\bar{F}_i = 1$ .
10:  end if
11: end for
12: for ( $k = 1$  to  $K$ ) do
13:    $\bar{\text{CCH}} = \text{CCH}$ .
14:    $\text{Tour}_k = [\text{Tour}_k, I]$ .
15:    $\text{visited}(k) = \phi$ .
16:    $\text{IoT\_list} = \mathcal{I}$ .
17:   Select an IoT device,  $r_k$ , randomly from  $\bar{\text{CCH}}$ .
18:    $\bar{\text{CCH}} = \bar{\text{CCH}} \setminus r_k$ .
19:    $J_k(r_k) = \text{IoT\_list} \setminus r_k$ .
20:    $\text{visited}(k) = [\text{visited}(k), r_k]$ .
21:    $\text{Tour}_k = [\text{Tour}_k, r_k]$ .
22:   Reorder the elements in  $\text{IoT\_list}$  randomly.
23:   for ( $i = 1$  to  $|\mathcal{I}|$ ) do
24:      $y = \text{IoT\_list}(i)$ .
25:     if [ $(y \in \text{visited}(k))$  OR  $(\epsilon < d_{yr_k})$ ] then
26:       Continue.
27:     else
28:        $\text{visited}(k) = [\text{visited}(k), y]$ .
29:       Add  $y$  to the cluster of the CH  $r_k$ .
30:     end if
31:     Select the next CH to be visited,  $s_k$ , from  $[(\mathcal{I} \setminus \text{visited}(k)) \cap$ 
        $\bar{\text{CCH}}]$  using (52) and (53).
32:      $\bar{\text{CCH}} = \bar{\text{CCH}} \setminus s_k$ .
33:      $\text{Tour}_k = [\text{Tour}_k, s_k]$ .
34:      $\text{visited}(k) = [\text{visited}(k), s_k]$ .
35:      $J_k(s_k) = J_k(r_k) \cup s_k$ .
36:      $r_k = s_k$ .
37:      $\tau(r_k, s_k) \leftarrow (1 - \omega_l) \tau(r_k, s_k) + \omega_l \tau_0$ .
38:     if ( $|\text{visited}(k)| == |\mathcal{I}|$ ) then
39:        $\text{Tour}_k = [\text{Tour}_k, F]$ .
40:       Break.
41:     end if
42:   end for
43: end for
44:  $\bar{T}_k = \sum_{(r,s) \in \text{Tour}_k} \Delta_1 + \bar{F}_s \Delta_2$ 
45: end for
46: Find the ant associated with the minimum tour cost, i.e.,  $k^* = \arg \min_k \bar{T}_k$ .
47:  $\text{Tour}^* = \text{Tour}_{k^*}$ 
48:  $\bar{T}^* = \bar{T}_{k^*}$ 
49: for (each edge  $(r, s) \in \text{Tour}_{k^*}$ ) do
50:    $\tau(r, s) \leftarrow (1 - \omega_g) \tau(r, s) + \omega_g \Delta \tau(r, s)$ .
51: end for
52:  $\text{Counter} = \text{Counter} + 1$ 
53: if ( $\text{Counter} < \text{Max\_Iteration}$ ) then
54:   Go to step 12.
55: end if

```

$\overline{\text{CCH}}$] to avoid reconsidering already visited nodes and CHs. Consequently, $\overline{\text{CCH}}$, Tour_k , $\text{visited}(k)$ and $J_k(s_k)$ are updated after adding a new CH to Tour_k . Then, the new selected CH, s_k , is set to be the current CH visited by ant k , i.e., $r_k = s_k$. Additionally, the pheromone over the edge between nodes r_k and s_k , $\tau(r_k, s_k)$, is update in line 37 as follows: $\tau(r_k, s_k) \leftarrow (1 - \omega_l) \tau(r_k, s_k) + \omega_l \tau_0$, where $0 < \omega_l < 1$ is a weighting parameter and τ_0 is a parameter corresponding to the initial pheromone concentration over any edge between two nodes. This step is called “the local-updating rule” in the ACS algorithm. Lines 38-41 aim in adding the final location F to Tour_k once all IoT devices are visited. The IoT device is considered visited if it is a CH or belonging to a cluster.

After all ants derived their tours, the objective function is calculated for all ants’ tours as shown in lines 44-46. Then, the best solution is derived in lines 47-49. For all edges corresponding the optimal tour, Tour_{k^*} , the pheromone over these edges is updated using the following rule:

$$\tau(r, s) \leftarrow (1 - \omega_g) \tau(r, s) + \omega_g \Delta\tau(r, s) \quad (54)$$

where $0 < \omega_g < 1$ is a weighting parameter, and $\Delta\tau(r, s)$ is express by

$$\Delta\tau(r, s) = \begin{cases} \frac{1}{\overline{T}^*}, & \text{if } (r, s) \in \text{Tour}^* \\ S, & \text{otherwise.} \end{cases} \quad (55)$$

where $\overline{T}^*$ is the objective function of the globally-best discovered tour, Tour^* , starting from the beginning of the trial. This updating rule is called “the global-updating rule” in the ACS algorithm. The iteration counter is incremented by one in line 53, then AC-JRC algorithm goes back to step 14 for further investigation for a better solution until the stopping condition is met. Finally, the AC-JRC algorithm provides a solution for IoT devices clustering and UAV trajectory path while considering the RIS and the jamming attack. Based on the procedures described in Algorithm 1, the time complexity of AC-JRC algorithm is $O(\text{Max_Iteration } K T^2)$.

V. SIMULATION RESULTS

We consider an area of $1500 \text{ m} \times 1500 \text{ m}$, where the UAV visits the CHs and collects the data of the IoT devices’ clusters. Since finding the optimal solution for our optimization problem is not an easy task when the number of variables is large, we consider a small scale problem when we compare the optimal solutions with our proposed algorithm. Moreover, larger scale scenarios are considered and the solutions are derived using our proposed algorithm. Accordingly, we divide the area into 25 and 100 possible locations for the IoT devices in the small and large scale scenarios, respectively. Additionally, the number of IoT devices are up to 20 and 80 in the small and large scale scenarios, respectively. The coordinates of the UAV’s initial and final locations are $(0, 0, 0)$ and $(1500, 1500, 0)$, respectively. On the other hand, the locations of the IoT devices, jammer and RIS are assigned randomly during each scenario. The results shown in this section are the

TABLE 3. Simulation parameters.

Parameter	Value
σ^2	-110 dBm
P	200 mW
P_J	0-6 W
$\bar{\gamma}$	20
d	0.1
μ	10^{-3}
λ	0.2
δ	10 s
S	39 m/s
δ	12 s
H	30 m
M_x	10
M_z	10
M	100
ϵ	250 m, 450 m
x^{max}	1500 m
y^{max}	1500 m
Max_Iteration	100
K	50
ζ	1
ω_l	0.1
ω_g	0.1
q_0	0.5

average of 30 random scenarios. Unless specified otherwise, the rest of the simulation parameters are described in Table 3.

CPLEX solver [20] over General Algebraic Modeling System (GAMS) [21] is utilized to derive the optimal solution of the ILP optimization problem. The CPLEX solver can derive the optimal solution of the ILP optimization problem using the Branch-and-cut method [22]. Additionally, MATLAB [23] computing platform is used to implement and simulate AC-JRC algorithm and the other benchmarking algorithms. The simulation is performed while using the Intel® Core™ i7-7500U processor with 8 GB of RAM.

In this section, the proposed AC-JRC algorithm is compared with three benchmarks: 1) The optimal solution, 2) AC-kmeans algorithm and 3) The standard Ant colony System (ACS) algorithm [15], [19]. The optimal solution provides a lower bound that helps us in studying how close the performance of AC-JRC algorithm is to the lower bound of the optimization problem. Since AC-JRC algorithm is based on Ant Colony System algorithm, AC-JRC algorithm is compared with the standard ACS algorithm. On the other hand, AC-JRC algorithm considers jointly the routing of the UAV (i.e., the ants in ACS) and the clustering of the IoT devices. To study the impact of the joint routing and clustering problem on the performance, AC-JRC algorithm is compared with what we call it AC-kmeans algorithm. AC-kmeans algorithm separately clusters the IoT devices and routes the UAV. Specifically, AC-kmeans algorithm clusters the IoT devices using a well-known clustering method called k-means clustering [24]. Additionally, AC-kmeans algorithm routes the UAV using ACS algorithm. The original Ant Colony System (ACS) algorithm solves the traveling salesman problem without considering the clustering problem. We slightly modified it to start from location I and finish at

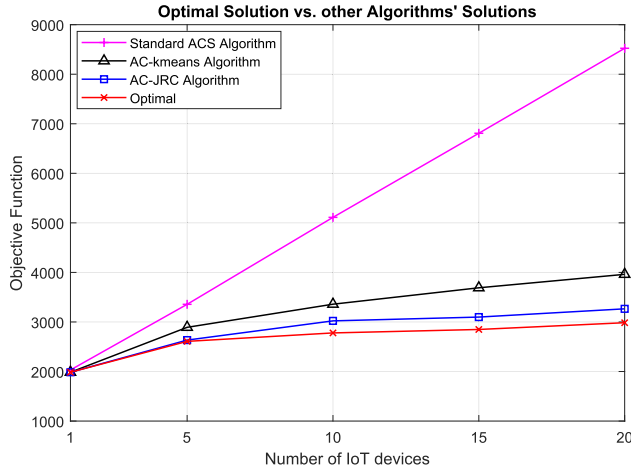


FIGURE 12. Comparing the performance of AC-JRC algorithm with the optimal solution and other benchmarks.

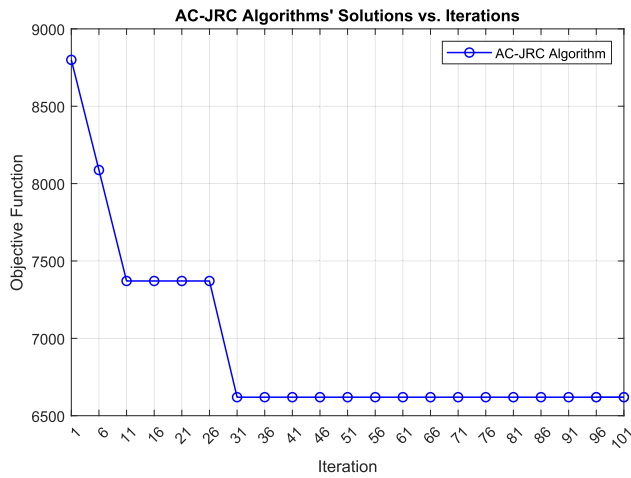


FIGURE 13. Convergence of AC-JRC Algorithm. The number of IoT devices and landing platforms are 50 and 20, respectively, $\epsilon = 250$ m and $P_J = 2W$.

location F by introducing a dummy node connecting I and F with zero costs to/from these two locations.

Fig. 12 shows the performance of our proposed algorithm, AC-JRC, compared with the optimal solution, K-means-based Ant Colony System algorithm (AC-kmeans) and the standard Ant Colony System (ACS) algorithm [15], [19]. We assume that the number of landing platforms is 10, $\epsilon = 250$ m and $P_J = 1W$. It is shown that the solution of AC-JRC is within 9% of the optimal solution and improves the performance of AC-kmeans and ACS algorithms by 17% and 61%, respectively, when the number of IoT devices is 20. Furthermore, AC-JRC significantly outperforms the original ACS algorithm and the ACS with k-means clustering algorithm (AC-kmeans algorithm). Additionally, the performance gap between the optimal solution and the two benchmark algorithms (AC-kmeans and ACS algorithms) becomes larger with increasing the number of IoT devices. In contrast to ACS and AC-kmeans algorithms, the gap between the performance of AC-JRC algorithm and the optimal solution is almost constant when the number of IoT devices is 10 and higher.

TABLE 4. Computational time (seconds).

Algorithm	Computational Time
AC-kmeans	38.4 s
AC-JRC	40.3 s
Standard ACS	66.7 s

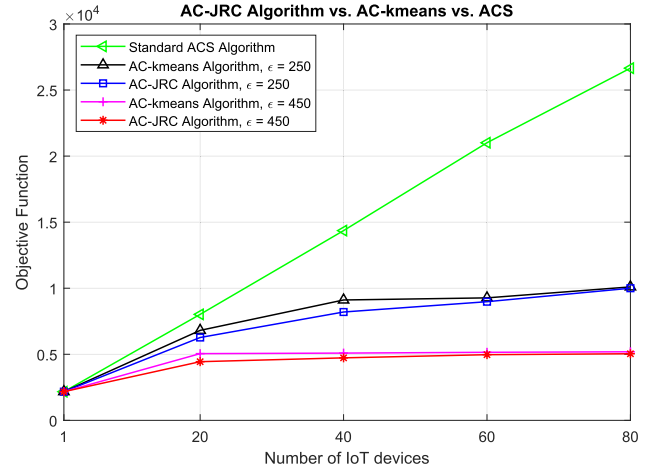


FIGURE 14. Effect of clustering and cluster radius to the value of the objective functions.

Fig. 13 presents the solutions of the AC-JRC algorithm during the first 100 iterations. By the end of each iteration of the AC-JRC algorithm, several parameters are updated including the pheromone vectors as described in Section IV. These updates can help the AC-JRC to derive better solutions during its iterations. It is shown how the AC-JRC algorithm enhances its performance significantly during its first 30 iterations. Then, it converges to a solution along the iterations from 30 to 100.

Table 4 shows the computational times of the AC-JRC, AC-kmeans and ACS algorithms. Both AC-JRC and AC-kmeans algorithms significantly outperform ACS algorithm in terms of computational time. The reason is that AC-JRC and AC-kmeans algorithms perform clustering for the IoT devices, then they utilize the idea of ACS algorithm in a subset of IoT devices (CHs). Hence, the computational time is much less than the time needed by ACS. It is shown in Table 4 that the computational times for AC-JRC and AC-kmeans are very close. Although AC-kmeans slightly outperforms the performance of AC-JRC in terms of computational time, AC-JRC outperforms AC-kmeans in terms of closeness to the optimal solution as depicted in Fig. 12.

Fig. 14 and Fig. 15 present the impact of the clustering and the cluster radius on the objective function and the number of used LPs. To study the impact of the clustering, we compare the performance of ACS with the performance of both AC-JRC and AC-kmeans algorithms. We also investigate the impact of joint UAV trajectory planning and IoT devices clustering on the performance by comparing the AC-JRC algorithm with the AC-kmeans algorithm. Additionally, Fig. 14 and Fig. 15 show the effect of the cluster radius on the objective function and the number of used LPs, respectively. We assumed here that $P_J = 4W$, and each IoT device

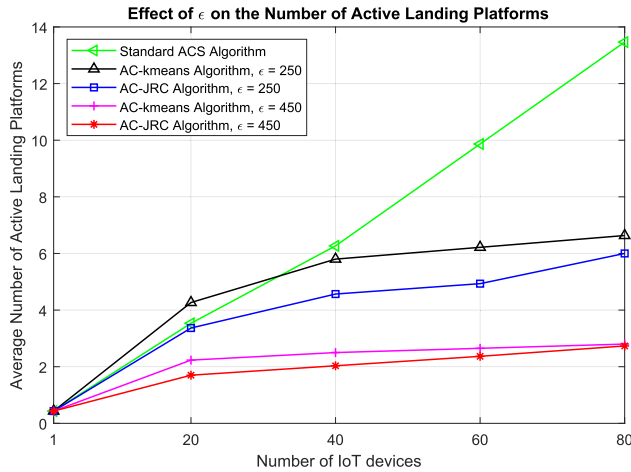


FIGURE 15. Effect of clustering and cluster radius to the number of used landing platforms.

is connected to an LP. By comparing the performance of the ACS with the AC-kmeans and AC-JRC, it is shown in Fig. 14 that clustering the IoT devices before collecting the data from them significantly decreases the objective function value. Moreover, Fig. 15 shows that clustering the IoT devices results in a lower number of active LPs except when the AC-kmeans is used and the number of IoT devices is approximately less than 30. However, our proposed algorithm AC-JRC outperforms both ACS and AC-kmeans in terms of the objective function value and the number of active LPs.

It is shown in Fig. 14 that AC-JRC algorithm outperforms AC-kmeans algorithm regardless of the number of IoT devices and cluster radius. When the cluster radius is larger, more IoT devices will be assigned to each cluster in general. Hence, the UAV tends to collect data from a fewer number of CHs. Accordingly, the average value for the objective function decreases when the cluster radius increases. Although increasing the radius reduces the value of the objective function, this comes at the cost of increasing the transmission range for the IoT devices to their cluster heads. Therefore, the cluster radius should be selected carefully to balance the contradicting goals of minimizing the UAV trajectory's distance and reducing the energy consumption of the IoT devices while sending their data to the CHs.

On the other hand, the performance gap between AC-JRC and AC-kmeans algorithms decreases when the number of IoT devices is large, as depicted in Fig. 14. Since there is a limit on the maximum number of clusters with a certain cluster radius within a certain area, no extra clusters are needed when the already created clusters cover the whole network area. Hence, the number of needed clusters does not increase significantly when the number of IoT devices is large and the already established clusters cover most of the network area. Therefore, the UAV flying trajectory tends to become slightly longer when the already created clusters cover most of the serving area. Accordingly, the performances of AC-JRC and AC-kmeans algorithms are close when the number of IoT devices is large.

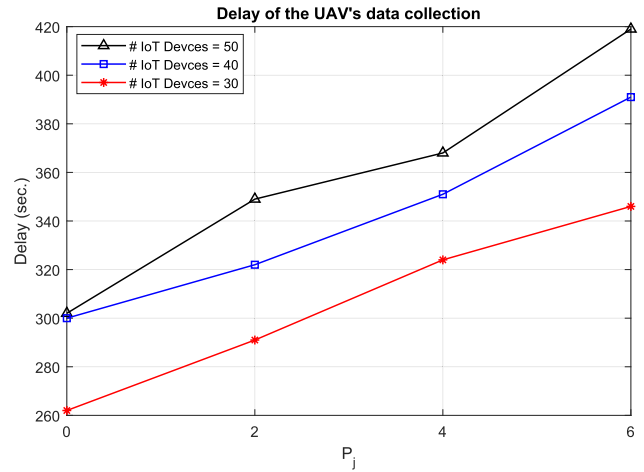


FIGURE 16. Delay of the UAV's data collection task.

Fig. 15 illustrates the relationship between the number of IoT devices, cluster radius and the number of used LPs. Larger radius tends to reduce the number of clusters and number of CHs accordingly. As the number of CHs decreases, the number of needed LPs tends to also decrease since the number of used LPs is upper bounded by the number of CHs. Therefore, the number of needed LPs decreases when the radius increases. On the other hand, a larger number of IoT devices tends to increase the number of clusters. As a result, the number of required CHs tends to increase. Since the jammer may block all CHs located within a certain area from uploading their data to the UAV through the wireless medium, only the IoT devices connected to LPs should be considered as CHs in the affected area. Increasing the number of served IoT devices increases the probability of the IoT device to be located within the jammer affected area. Therefore, increasing the number of IoT devices increases the number of used LP, as presented in Fig. 15.

In Fig. 16, the impact of the jamming power and the number of IoT devices on the delay of data collection process is investigated. We calculate the total traveled distance by the UAV starting from the initial location until the final location. Then, we derive the total delay given that the UAV's speed is S . Since increasing the number of IoT devices tends to increase the potential of having more CHs, the distance traveled by the UAV to collect data from these CHs tends to be higher when the number of IoT devices are higher. Hence, the delay increases when the UAV serves more IoT devices. On the other hand, a higher jamming power tends to impact a higher number of IoT devices and increase the chance that the UAV is blocked from collecting data from a larger number of IoT devices. Therefore, increasing the jamming power is narrowing the choices for the UAV for selecting CHs that results in the shortest flying trajectory. As a result, a higher jamming power tends to increase the UAV flying trajectory and delay.

Fig. 17 demonstrates the significance of utilizing the proposed landing platforms in the UAV data collection in the presence of a malicious jammer. The proposed scheme that

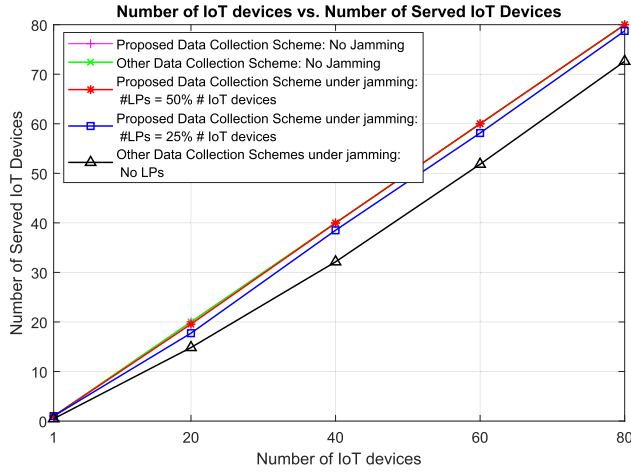


FIGURE 17. Comparing the proposed landing platform-based UAV data collection with the other schemes that do not utilize the landing platform.

utilizes the LPs aims in mitigating the effect of the jammer and serves the IoT devices even if they are within the jamming affected areas. Hence, we consider the number of served IoT devices as a key performance indicator when comparing the proposed network architecture with the other schemes that do not utilize the LPs. To the best of our knowledge, this paper is the first one that addresses this issue by utilizing the proposed LPs. Therefore, we consider the current classical way of dealing with jamming attacks without using LPs as the “Other Schemes”. Moreover, we investigate the number of served IoT devices with and without the presence of the jammer to see the impact of the LPs on the number of served IoT devices.

Fig. 17 presents how many IoT devices served by the UAV versus the total number of IoT devices in the networks. It is shown that all the IoT devices are served regardless of whether they utilize the LPs or not when there is no jammer (magenta and green curves, respectively). This is due to the fact that the UAV can hover over all CHs and collect the data from them without interference from the jammer. On the other hand, the number of served IoT devices decreases significantly when the LPs are not used (Other data collection schemes under jamming). For example, around 10% of the total IoT devices are not served when the LPs are not utilized and the number of IoT devices is 80. However, the proposed scheme significantly improves the number of served IoT devices under a jamming attack when only 25% of the IoT devices are connected to LPs (the blue curve). Additionally, all the IoT devices under jamming are served when 50% of the IoT devices are connected to LPs (the red curve). Interestingly, utilizing the LPs in the network architecture can mitigate or even eliminate the effect of the malicious jammer when enough LPs are adopted.

Fig. 18 illustrates how the jammer's transmission power impacts the value of the objective function. We assume here that $\epsilon = 250$ m and each IoT device is connected to an LP. As the value of P_J increases, a wider area will be impacted

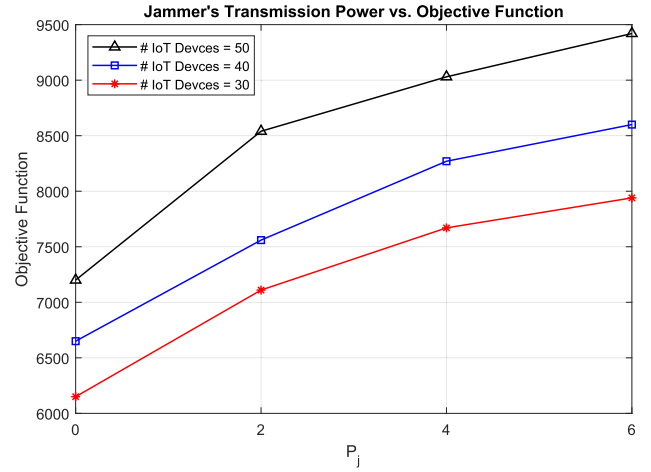


FIGURE 18. Effect of the jammer's transmission power on the objective function.

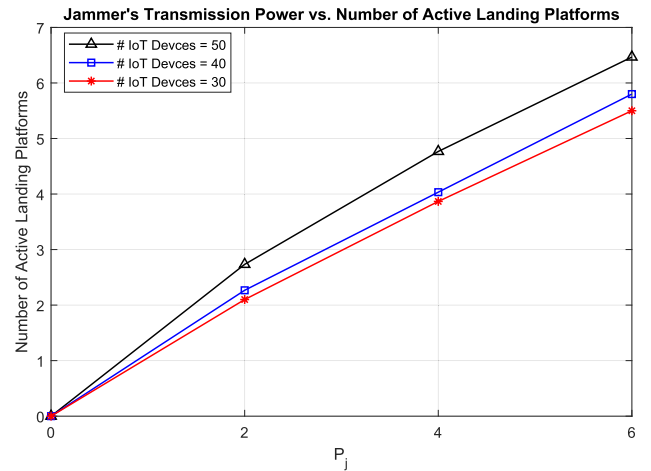


FIGURE 19. Effect of the jammer's transmission power on the number of used LPs.

by the jammer. Thus, the probability of having more CHs impacted by the jammer will be higher. As a result, the UAV may be obligated to take a longer flying path to collect data from another set of CHs that are not impacted by the jammer. On the other hand, Fig. 19 indicates that a higher jamming power results in more used LPs by the UAV. Hence, the UAV may need to land more LPs to collect data, and that causes extra costs to the objective function. Accordingly, increasing the jamming power impacts clustering and the number of used LPs, and consequently, increasing the value of the objective function. It is shown that all curves in Fig. 18 and Fig. 19 have similar increasing trends regardless of the number of IoT devices in the networks.

Fig. 20 presents the impact of the jamming power to the number of served IoT devices by the UAV while considering different values for the SINR threshold, $\bar{\gamma}$, and different values for the number of available LPs. The total number of IoT devices is 40 and the cluster radius of each cluster is 250 m. We assume that the IoT device is served by the UAV only when it is a member of the cluster. Hence, our goal is to collect the data from all IoT devices through the

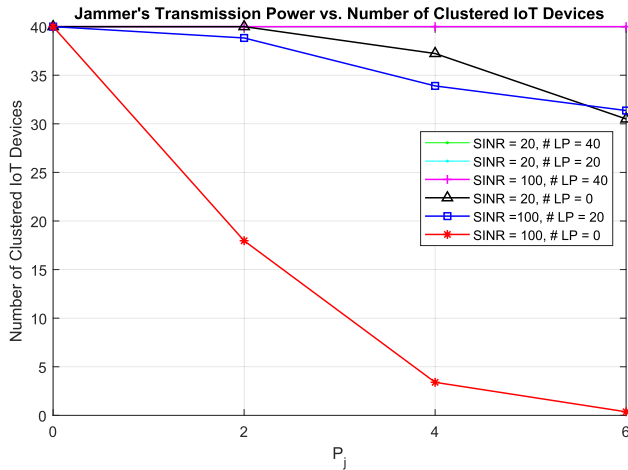


FIGURE 20. Number of served IoT devices vs. jammer's transmission power vs. SINR.

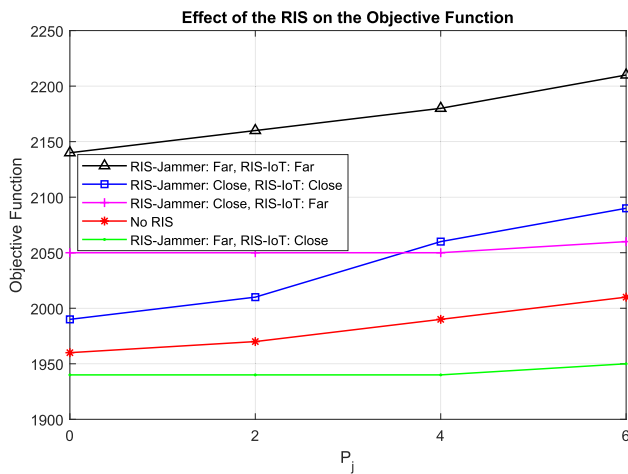


FIGURE 21. Effect of the RIS and its location on the objective function.

CHs or at least maximizing the number of clustered (served) IoT devices. Obviously, the number of clustered IoT devices declines as the jamming power and the SINR threshold increase in general. The exception for this decline in the number of served IoT devices is when each IoT device is connected to an LP (when the number of LPs is 40), where all IoT devices are served. This indicates that the LPs provide a robust data collection scheme against excessive jamming power. When we do not adopt the LPs in the network and the SINR threshold and the jamming power are high, no IoT device can be served by the UAV. In contrast, all IoT devices can be served regardless of the jamming power when enough LPs are added to the network. Given a certain SINR threshold, increasing the number of available LPs in the network causes the UAV to serve more IoT devices. When the jamming power is very high ($P_j = 6$ W), it is shown that the number of served IoT devices with 20 LPs and $\bar{\gamma} = 20$ is higher than the number of served IoT devices with no LPs and $\bar{\gamma} = 100$. Interestingly, we can consider a higher value for the SINR threshold while serving more IoT devices if enough LPs are adopted in the network.

Fig. 21 presents the impact of the RIS's location on the value of the objective function. For simplicity, we consider one IoT device and study the impact of the RIS's location on the objective function. We also consider five scenarios described as follows:

- The distances between the RIS and the jammer and between the RIS and the IoT devices are large (RIS-Jammer: Far, RIS-IoT: Far).
- The distances between the RIS and the jammer and between the RIS and the IoT devices are small (RIS-Jammer: close, RIS-IoT: close).
- The distances between the RIS and the jammer and between the RIS and the IoT devices are small and large, respectively (RIS-Jammer: close, RIS-IoT: far).
- The distances between the RIS and the jammer and between the RIS and the IoT devices are large and small, respectively (RIS-Jammer: far, RIS-IoT: close).
- No RIS: no RIS is considered in the network architecture.

It is shown that the RIS can cause a positive or a negative impact to the data collection process depending on its location and how close it is from the jammer and the IoT device. When the RIS is near the jammer, it may contribute to the jamming attack by reflecting the jammer's signal to the UAV. On the other hand, the RIS located nearby the IoT device can contribute positively in the data collection process by reflecting its signal to the UAV. Fig. 21 indicates that the best performance is achieved when the RIS is far from the jammer and close to the IoT device (RIS-Jammer: far, RIS-IoT: close). In this scenario, the RIS tends to only reflect the IoT device's signal. Hence, the UAV will have a better chance to collect the data from a bigger area and from a location that causes a shorter flying trajectory. On the other hand, the worst performance is achieved when the RIS is far from both the IoT device and the jammer but the IoT device is close to the jammer (RIS-Jammer: Far, RIS-IoT: Far). In this scenario, the RIS does not significantly impact the performance. However, the performance is the worst due to the fact that the IoT device and the jammer are close to each other.

When the RIS is close to the jammer but far away from the IoT device (RIS-Jammer: close, RIS-IoT: far), this implies that the IoT device is also far away from the jammer. Since the IoT device is far away from the RIS and the jammer, the performance does not change regardless of the increase in the jamming signal. However, the performance is significantly impacted by the jamming signal when both the RIS and the jammer are close to the IoT device (RIS-Jammer: close, RIS-IoT: close). Interestingly, the scenario where no RIS is used outperforms all scenarios except the scenario where the RIS is far away from the jammer and close to the IoT device. This is due to the fact that the jammer can utilize the RIS to deteriorate the performance when the place of the RIS is not optimal. Accordingly, the RIS should be placed carefully in the network when there is a possibility of jamming attack

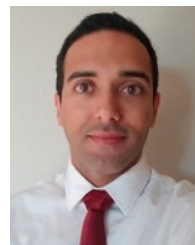
to get the advantages from it and avoid its disadvantage of reflecting the jamming signal.

VI. CONCLUSION

This paper investigated the data collection problem in IoT using a UAV in the presence of a jamming signal. We introduced a novel network component called a Landing Platform, which is connected to an IoT device physically and provides an alternative and resilient communication method when the wireless medium is inaccessible due to a high jamming power. Furthermore, we employed an RIS which assists the IoT devices in reflecting their signals to the UAV when the indirect channel to the UAV through the RIS is better. Clustering of the IoT devices and designing the UAV's path are jointly optimized, where we formulated our problem mathematically. Then, we proposed the AC-JRC algorithm, which is Ant Colony-based and solves the optimization problem in a more efficient way. Simulation results showed how the proposed scheme improves the performance and enhances the network defense against jamming attacks. We evaluated the performance of the AC-JRC algorithm and investigated the positive impact of adopting the LPs in the network and how the location of the RIS critically impacts the performance.

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ABDULLAH M. ALMASOUD received the B.Sc. degree in computer engineering from King Saud University, Riyadh, Saudi Arabia, and the M.Sc. degree in computer engineering and the Ph.D. degree in computer engineering and electrical engineering from Iowa State University, Ames, IA, USA. He is currently an Assistant Professor with the Department of Electrical Engineering, Prince Sattam Bin Abdulaziz University, Al-Kharj, Saudi Arabia. His research interests include wireless networks, cognitive radio networks, the Internet of Things, UAV-assisted networking, and RF energy harvesting. He was a recipient of the Best Paper Award at the IEEE Global Communications Conference 2018 on Ad Hoc and Sensors Networks Symposium.

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